BATTERY REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

id0117008005q7

Operation After Replacing Battery (With i-stop)

Note

- If the battery charge/discharge flow (number of times battery can operate) exceeds the specification, a DTC is recorded and the i-stop warning light (amber) flashes. If the battery charge/discharge flow (number of times battery can operate) exceeds the specification, replace the battery with a new one.
- When replacing the battery, perform the following procedure: When replacing the PCM simultaneously, perform the PCM configuration first. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See PCM CONFIGURATION (USING READ/WRITE FUNCTION) [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See PCM CONFIGURATION (USING AS-BUILT DATA) [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

Step	Action
1	Switch the ignition ON (engine off).
2	Shift the selector lever to the N position. (ATX) Shift the shift lever to the neutral position. (MTX)
3	Perform the following work with the brake pedal depressed. 1. Depress the accelerator pedal for 5 s or more. 2. Verify that the charging system warning light and the master warning light flash. 3. Depress and release the accelerator pedal 3 times. 4. Verify that the charging system warning light illuminates and the master warning light turns off.
4	Switch the ignition off and disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
5	Verifying battery condition initialization setting (i-stop setting). (See BATTERY CONDITION INITIALIZATION SETTING (i-stop SETTING).)

Operation After Replacing Battery (with i-ELOOP, without i-stop)

Note

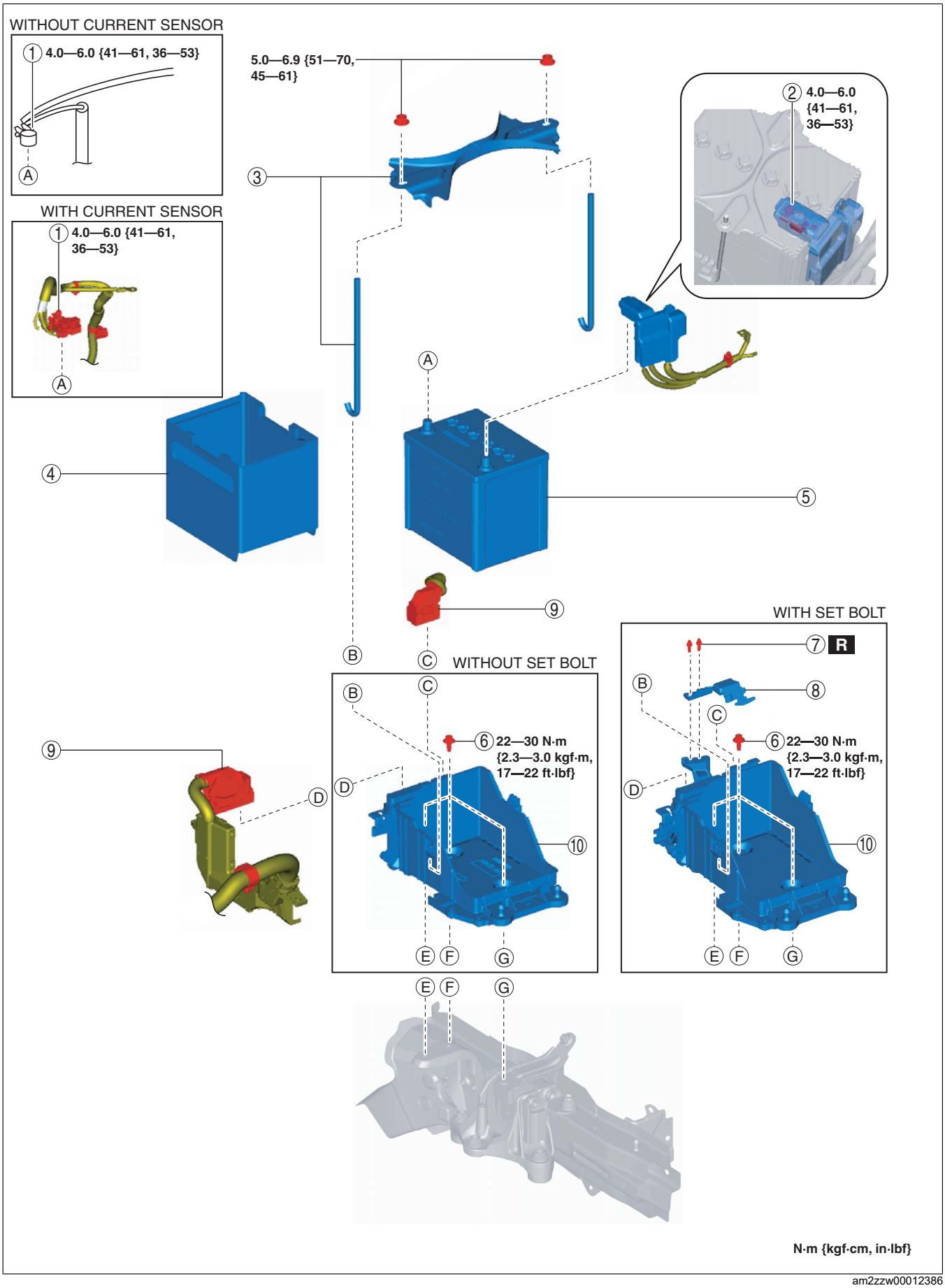
- If the battery charge/discharge flow (number of times battery can operate) exceeds the specification, a DTC is recorded and the master warning light flashes. If the battery charge/discharge flow (number of times battery can operate) exceeds the specification, replace the battery with a new one.
- When replacing the battery, perform the following procedure: When replacing the PCM simultaneously, perform the PCM configuration first. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See PCM CONFIGURATION (USING READ/WRITE FUNCTION) [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See PCM CONFIGURATION (USING AS-BUILT DATA) [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)

Step	Action
1	Switch the ignition ON (engine off).
2	Shift the selector lever to the N position.
3	Perform the following work with the brake pedal depressed. 1. Depress the accelerator pedal for 5 s or more. 2. Verify that the charging system warning light and the master warning light flash. 3. Depress and release the accelerator pedal 3 times. 4. Verify that the charging system warning light illuminates and the master warning light turns off.
4	Switch the ignition off and disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)

Battery Removal/Installation

1. Remove in the order indicated in the table.
2. Install in the reverse order of removal.

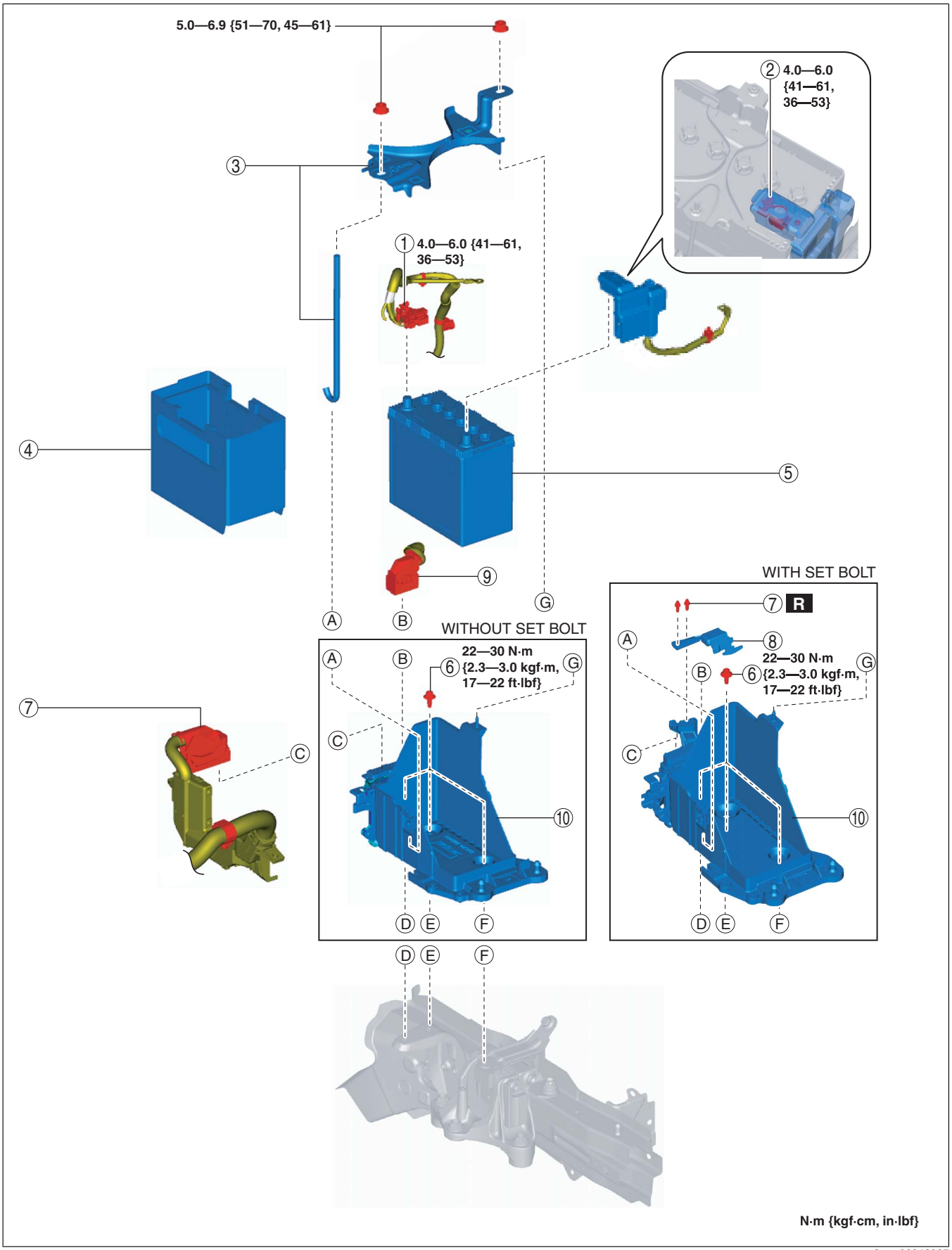
Without i-ELOOP



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1	Negative battery cable (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) (See Negative battery cable connection note (with current sensor) .)
2	Positive battery cable
3	Battery clamp
4	Battery box (See Battery box installation note.)
5	Battery
6	Battery tray bolt
7	Set bolt (With set bolt) (See Battery tray and PCM component removal note.) (See Set bolt installation note.)
8	PCM cover (With set bolt) (See Battery tray and PCM component removal note.)
9	PCM connector (See Battery tray and PCM component removal note.) (See PCM connector connection note.)
10	Battery tray and PCM component (See Battery tray and PCM component removal note.)

With i-ELOOP



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1	Negative battery cable (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) (See Negative battery cable connection note (with current sensor) .)

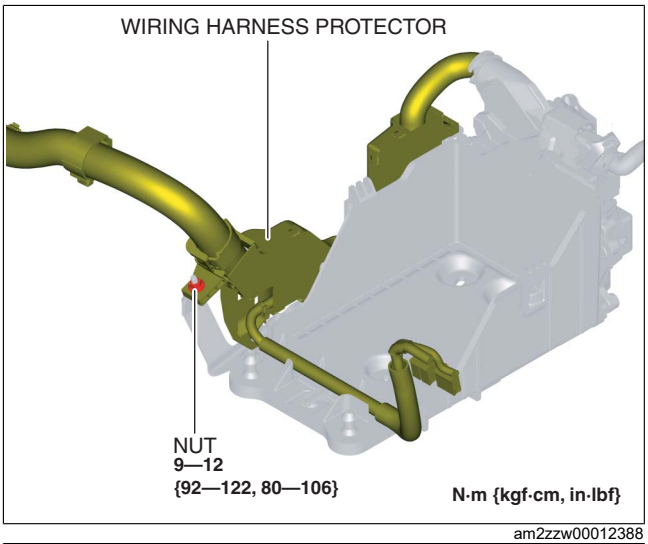
2	Positive battery cable
3	Battery clamp

4	Battery box (See Battery box installation note.)
5	Battery
6	Battery tray bolt
7	Set bolt (With set bolt) (See Battery tray and PCM component removal note.) (See Set bolt installation note.)
8	PCM cover (With set bolt) (See Battery tray and PCM component removal note.)

9	PCM connector (See Battery tray and PCM component removal note.) (See PCM connector connection note.)
10	Battery tray and PCM component (See Battery tray and PCM component removal note.)

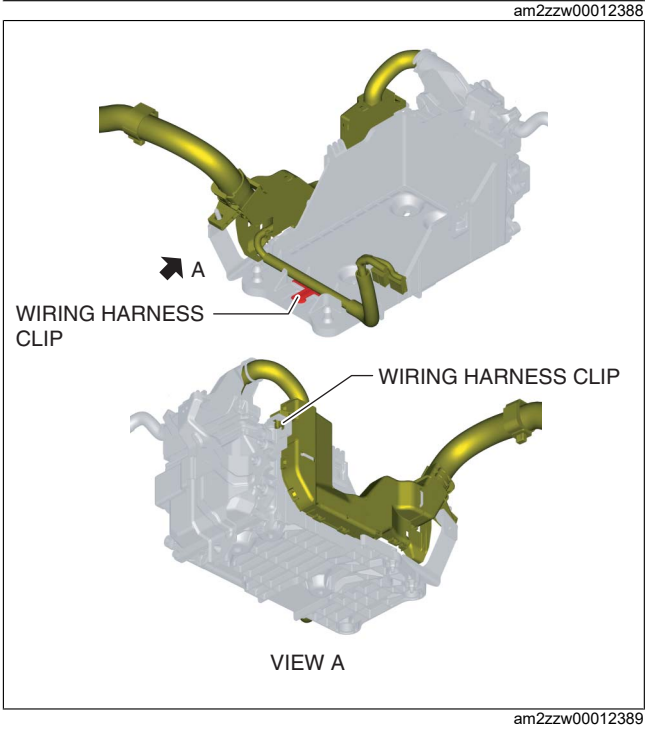
Battery tray and PCM component removal note

1. Remove the air cleaner, air hose and fresh air duct component. (See INTAKE-AIR SYSTEM REMOVAL/ INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
2. Remove the nut securing the wiring harness protector to the battery tray.



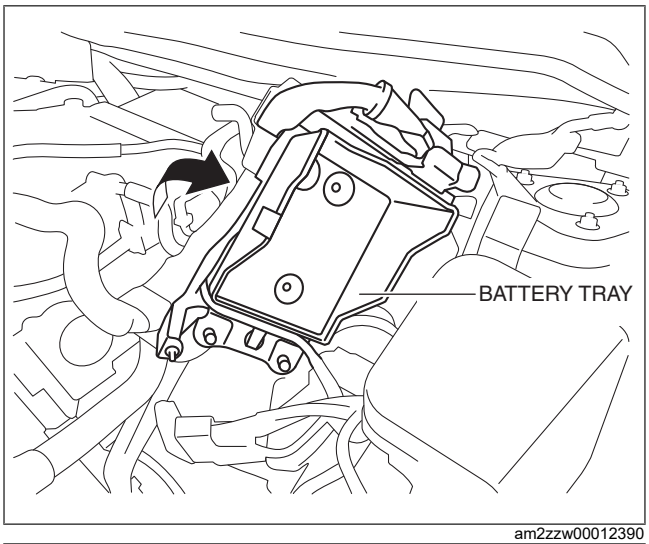
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3. Release the wiring harness clips in the two locations shown in the figure.
4. Disconnect the service plug. (with i-ELOOP) (See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].)
5. DC-DC converter (i-ELOOP) (with i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/ INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].)



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6. Pull out the battery tray and PCM component to the position shown in the figure to remove the set bolt. (With set bolt)

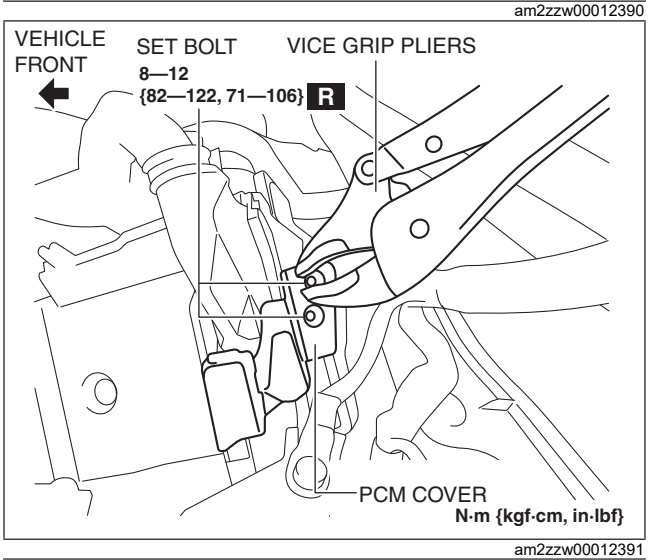


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7. Remove the two set bolts shown in the figure using a pair of vice grip pliers. (With set bolt)

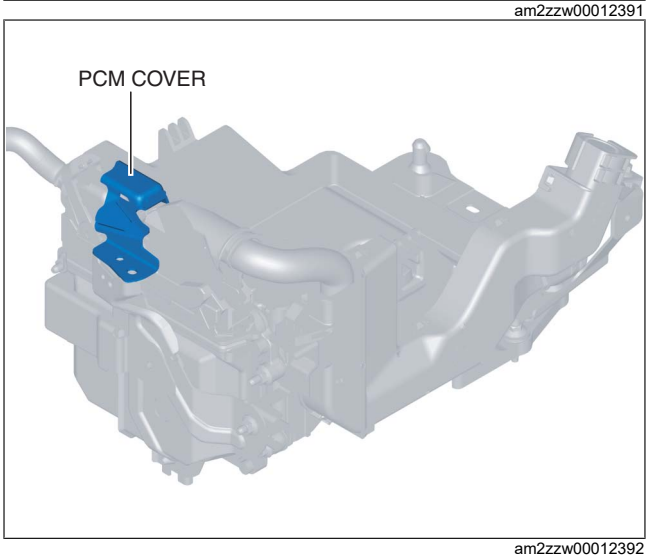
Warning

- **The vice grip pliers could slip on the conically-shaped set bolts and cause injury. Keep the set bolts wedged firmly in the vice grip pliers so that the pliers do not slip.**



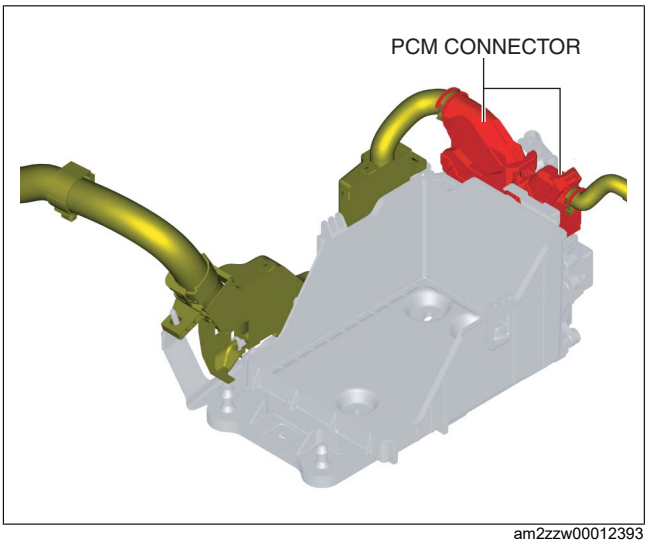
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8. Remove the PCM cover shown in the figure. (with set bolt)



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9. Disconnect the PCM connector shown in the figure.
10. Remove the battery tray and PCM component.

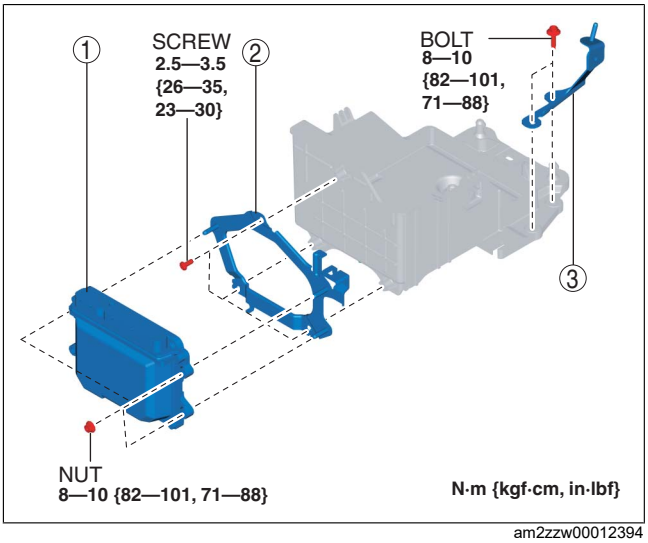


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11. When replacing the battery tray with a new one, perform the following procedure:

Without set bolt

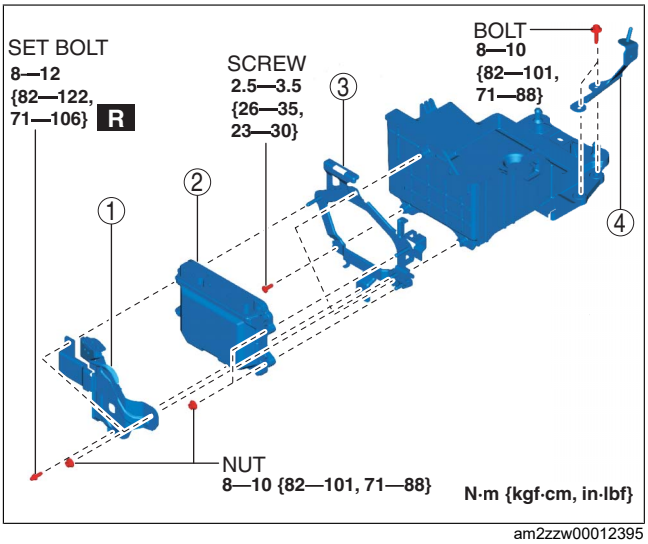
1. Remove the PCM.
2. Remove the bracket.
3. Remove the bracket.



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With set bolt

1. Remove the bracket.
2. Remove the PCM.
3. Remove the bracket.
4. Remove the bracket.



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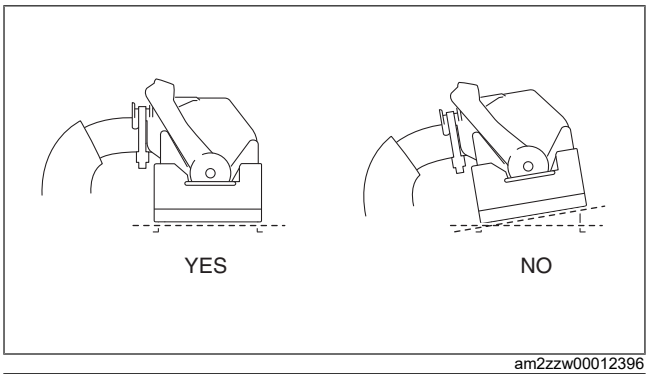
PCM connector connection note

Caution

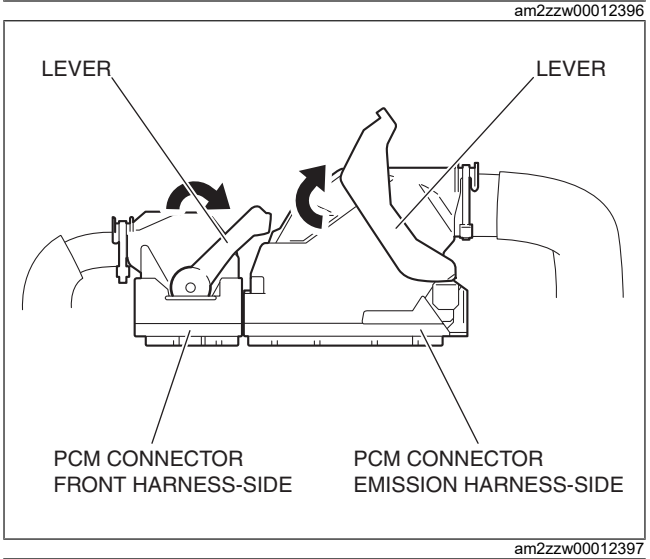
- **Do not touch the PCM connector terminal. The terminal is extremely thin and can be damaged by touching it.**

- **If the PCM connector is inserted at an angle and the lever is moved, the connector could be damaged. Verify that the PCM connector is inserted straight.**

1. Set the PCM connector lever to the position shown in the figure.

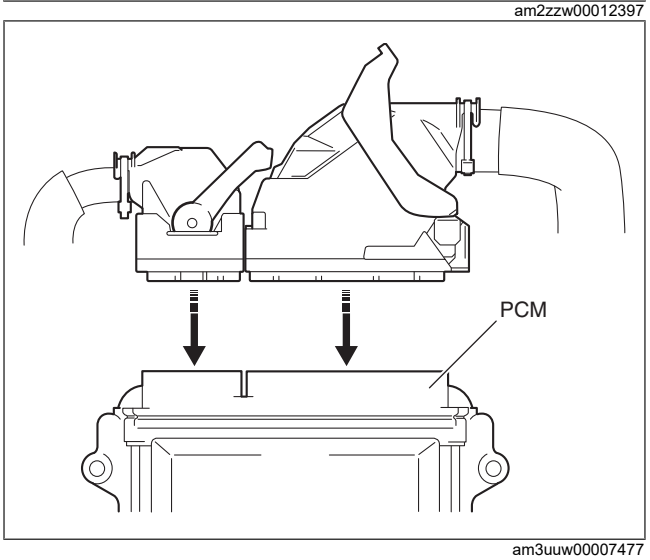


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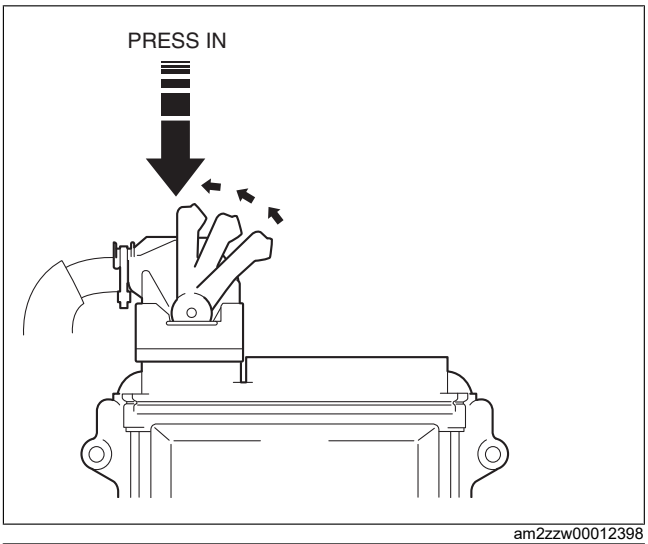
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2. Align the PCM connector straight against the connection surface.



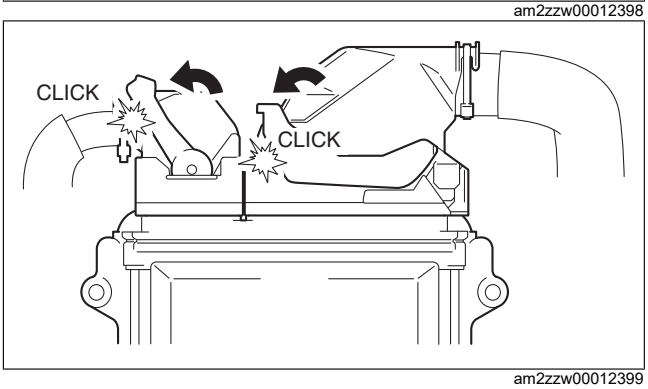
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3. Insert the PCM connector straight and press it in until the lever moves up naturally. (Front harness-side connector)



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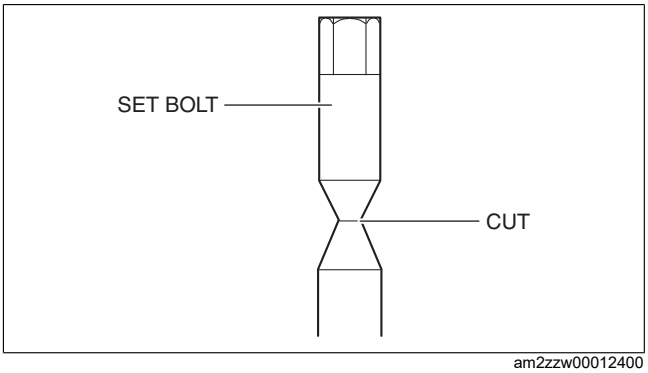
4. Press the PCM connector lever until a click sound is heard.



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Set bolt installation note

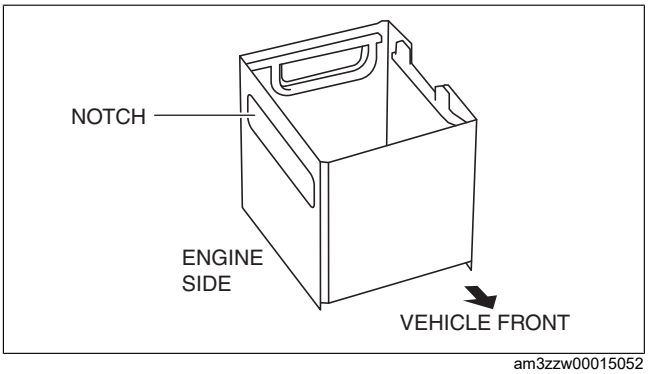
1. Install a new set bolt and tighten it until the neck of the bolt is cut.



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Battery box installation note

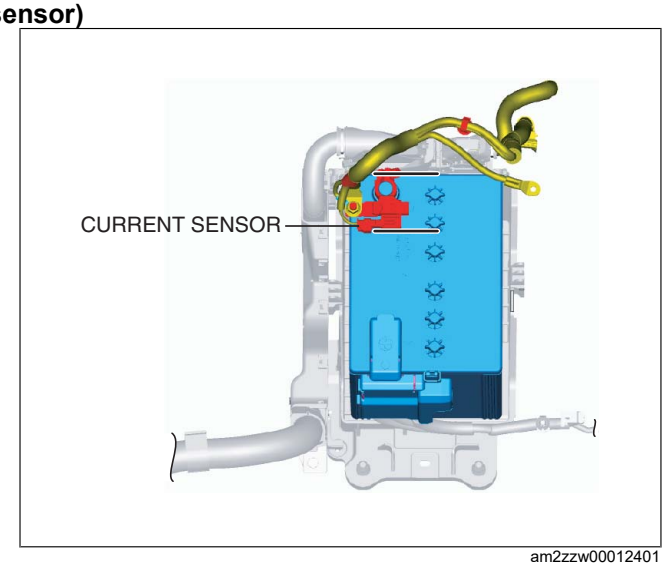
1. Install the battery box so that the side with the larger notch is pointed at the engine side.



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Negative battery cable connection note (with current sensor)

1. Connect the negative battery cable with the current sensor and the battery parallel to each other.



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